

# Brain Tumor Classification Using Deep Learning

## Introduction

Brain tumor classification is a critical task in medical diagnostics, enabling early detection and effective treatment. Brain tumors can be of different types, such as **Glioma, Meningioma, and Pituitary tumors**, or there may be no tumor at all (**No Tumor category**). Manual classification of brain MRI scans is time-consuming and prone to errors. With the advancement of deep learning techniques, particularly **Convolutional Neural Networks (CNNs)**, automated classification of brain tumors has become significantly more accurate and reliable.

This project implements a **VGG-based CNN model** to classify brain MRI scans into four categories. Additionally, a **Graphical User Interface (GUI)** is developed using **PyQt5**, allowing users to upload MRI images and obtain classification results efficiently.

## Research Paper

The methodology implemented in this project is inspired by the research paper "**VGG-C Transform with Batch Normalization for MRI Classification**". The paper explores the effectiveness of deep learning in classifying MRI scans using **VGG-based CNN architectures**. This research provided the foundation for the model architecture, training methodology, and classification techniques used in this project.

## Methodology

### 1. Dataset

The dataset used for training and evaluation is from **Kaggle** :

<https://www.kaggle.com/datasets/navoneel/brain-mri-images-for-brain-tumor-detection> . It contains MRI scans classified into four categories:

- **Glioma Tumor**
- **Meningioma Tumor**
- **No Tumor**
- **Pituitary Tumor**

The dataset is preprocessed to ensure uniformity and compatibility with the deep learning model.

## 2. Data Preprocessing

To prepare the MRI images for training, the following preprocessing steps are applied:

- **Grayscale Conversion:** Since MRI scans are already grayscale, this step ensures uniform input.
- **Resizing:** Images are resized to **208×208 pixels** to maintain consistency.
- **Normalization:** Pixel values are normalized using **mean=0.5** and **standard deviation=0.5** to enhance training stability.
- **Data Augmentation:** Techniques such as flipping and rotation can be added in future versions to improve model generalization.

## 3. Model Architecture

The implemented model is based on **VGG-C Transform with Batch Normalization**, consisting of:

- **Convolutional Layers:** Extracts key spatial features from the MRI images.
- **Batch Normalization:** Speeds up training and improves generalization.
- **Max-Pooling Layers:** Reduces dimensionality while retaining important features.
- **Fully Connected Layers:** Classifies the image into one of the four categories.

## 4. Training Methodology

- **Loss Function:** **CrossEntropyLoss** is used to handle the multi-class classification problem.
- **Optimizer:** The **Adam optimizer** is used with a learning rate of **0.001**.
- **Epochs:** The model is trained for **10 epochs** to ensure optimal learning.
- **Batch Size:** A batch size of **32** is used to optimize GPU utilization.

## 5. Graphical User Interface (GUI)

A user-friendly **GUI** is implemented using **PyQt5** to enhance usability. Features include:

- **Upload Image Button:** Allows users to upload an MRI scan.

- **Prediction Display:** Shows the detected tumor classification.
- **Confidence Score Graph:** A bar chart displays confidence scores for all classes.

## Evaluation Matrix

To assess model performance, the following evaluation metrics are considered:

### 1. Loss Curve

- The model's training loss is plotted over epochs to monitor convergence.
- A gradual decrease in loss indicates proper learning.

### 2. Accuracy Score

- The classification accuracy is evaluated using test data.
- Accuracy provides insights into the model's generalization ability.

### 3. Confidence Scores Visualization

- A bar chart displays the confidence levels for each classification.
- Higher confidence in correct predictions indicates a well-trained model.

## Results

The model was trained for **10 epochs**, and the training loss values observed over epochs were:

Epoch [1/10], Train Loss: 5.4790

Epoch [2/10], Train Loss: 1.0116

Epoch [3/10], Train Loss: 1.0314

Epoch [4/10], Train Loss: 1.1855

Epoch [5/10], Train Loss: 1.2172

Epoch [6/10], Train Loss: 1.1903

Epoch [7/10], Train Loss: 1.0299

Epoch [8/10], Train Loss: 0.9601

Epoch [9/10], Train Loss: 0.9461

Epoch [10/10], Train Loss: 0.9172

The gradual reduction in training loss indicates that the model successfully learned the underlying patterns in the MRI dataset. The final trained model is saved at: **E:/Intern/MEGA-MINDS/Trained Model/model.pth**

## Conclusion and Future Scope

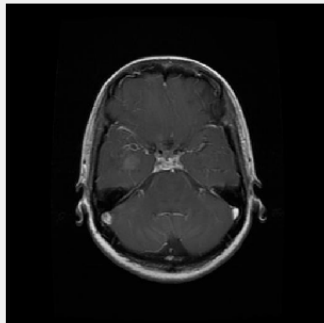
This project successfully classifies brain MRI images into **four tumor categories** using deep learning. The integration of **VGG-based CNN models** with a **PyQt5 GUI** enhances usability, making the system accessible to medical professionals.

## Future Improvements

- **Dataset Expansion:** Training on a larger and more diverse dataset to improve accuracy.
- **Advanced Architectures:** Experimenting with **ResNet**, **EfficientNet**, or **Transformers** for enhanced performance.
- **Real-Time MRI Processing:** Integrating with hospital systems for real-time analysis.
- **Explainability Features:** Adding **Grad-CAM visualization** to interpret model decisions.

This research serves as a significant step toward **AI-assisted medical diagnostics**, helping radiologists detect and classify brain tumors efficiently and accurately.

```
PS E:\Intern\MEGA-MINDS> & "C:/Users/Himanshu Gaur/AppData/Local/Programs/Python/Python39-64/Scripts/python.exe" E:\Intern\MEGA-MINDS\train.py
Epoch [1/10], Train Loss: 5.4790
Epoch [2/10], Train Loss: 1.0116
Epoch [3/10], Train Loss: 1.0314
Epoch [4/10], Train Loss: 1.1855
Epoch [5/10], Train Loss: 1.2172
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Epoch [8/10], Train Loss: 0.9601
Epoch [9/10], Train Loss: 0.9461
Epoch [10/10], Train Loss: 0.9172
Model saved to E:/Intern/MEGA-MINDS/Trained Model/model.pth
PS E:\Intern\MEGA-MINDS> █
```



Upload Image

Predict

Prediction: glioma

